VisioSphere — Detection Algorithms: Significant Calculations

Methodology extract from cctv_core.py, live production configuration as of 9 September 2026. All formulas as implemented. Angles in degrees; times in seconds; image coordinates with the y-axis increasing downward.

Notation

Symbol	Definition	Symbol	Definition
s_{mid}	Shoulder midpoint, mean of keypoints 5 and 6	h_{bb}, w_{bb}	Bounding-box height and width, pixels
h_{mid}	Hip midpoint, mean of keypoints 11 and 12	θ	Torso inclination from vertical, degrees
L_t	Torso length, aspect-corrected (Eq. 4)	α	Bounding-box aspect ratio, h_{bb}/w_{bb}
w_s	Shoulder width, $\lvert kp_5 - kp_6 \rvert$	κ	Pixel-aspect correction factor (Eq. 2)
c	Per-keypoint detection confidence, 0–1	Δt	Inter-frame interval, seconds
φ	Joint flexion angle, degrees (Eq. 9)	Ω	Oscillation term, agitation (Eq. 20)

1. Preprocessing common to all detectors

1.1 Keypoint confidence gating

A person instance is admitted to geometric analysis only if the pose detector returns a person-level confidence of at least 0.10 and at least 5 of the 17 COCO keypoints exceed a confidence of 0.45. Individual keypoints below 0.45 are excluded from every measurement in which they would otherwise participate.

1.2 Temporal smoothing of keypoint coordinates

Per-frame pose estimates carry high-frequency positional error. Each admitted keypoint is smoothed by a first-order exponential moving average:

$$x_t = 0.40 \cdot x_t^{raw} + 0.60 \cdot x_{t-1} \tag{1}$$

where x_{t-1} is the smoothed value from the previous frame. When $c < 0.15$ the coordinate is held at its previous value rather than updated, preventing a low-confidence estimate from displacing a stable track.

1.3 Pixel-aspect correction

Inference is performed at 640 × 360 on a source stream of differing aspect ratio. The anisotropic resize scales the horizontal and vertical axes unequally, so an angle measured directly from resized pixel coordinates is systematically biased. A correction factor is applied to every angular measurement:

$$\kappa = (W_{src} / H_{src}) \div (W_{inf} / H_{inf}) \tag{2}$$

$$\tan \theta_{true} = \kappa \cdot \tan \theta_{measured} \tag{3}$$

where W_{src}, H_{src} are the source frame dimensions and W_{inf}, H_{inf} the inference dimensions. Horizontal displacements entering a length or angle calculation are likewise scaled by κ.

1.4 Torso length and anthropometric plausibility

Torso length is the reference scale for every normalised measurement in this document. It is computed with the horizontal component aspect-corrected:

$$L_t = \sqrt{\left[\left((s_{mid,x} - h_{mid,x}) \cdot \kappa \right)^2 + \left(s_{mid,y} - h_{mid,y} \right)^2 \right]} \tag{4}$$

A candidate whose torso is implausibly short relative to its shoulder width is a spurious skeleton fitted to background structure. Such instances are rejected before any state evaluation:

$$L_t / w_s \geq 0.60 \tag{5}$$

1.5 Identity persistence

Person instances are associated across frames by ByteTrack, using two-stage association on detection score: high-confidence detections (≥ 0.15) are matched first, then remaining tracks are matched against low-confidence detections (≥ 0.05), which recovers subjects partially occluded or mid-transition. New tracks initialise at 0.10; lost tracks are retained for a 90-frame buffer; the association threshold is an IoU of 0.85. Each state machine described below is instantiated per track.

2. Postural geometry — shared by fall and lying-down detection

Falling and voluntarily lying down terminate in the same posture and are therefore governed by a single state machine. The measurements in this section establish that a subject is in a fallen state; Section 3 determines whether the transition into that state was a fall.

2.1 Torso inclination

$$\theta = \left| \text{degrees}\left(\text{atan2}(\Delta x \cdot \kappa , \Delta y) \right) \right|, \quad \Delta = s_{mid} - h_{mid} \tag{6}$$

where θ = 0° denotes an upright trunk and θ = 90° a horizontal trunk.

2.2 Bounding-box aspect ratio

$$\alpha = h_{bb} \,/\, w_{bb} \tag{7}$$

where $\alpha > 1.5$ is characteristic of an upright subject and $\alpha < 0.9$ of a subject on the ground. Being derived from the enclosing box rather than the skeleton, α remains available when keypoints are partially occluded.

2.3 Height collapse ratio

Absolute pixel height varies with the subject's distance from the camera. The collapse ratio expresses current height against that track's own observed maximum, the reference being updated only on frames classified as upright:

$$r_h = h_{bb} \,/\, \max(\, h_{bb} \,)_{\text{upright}} \tag{8}$$

2.4 Body extent ratio

Dividing the bounding-box height by the subject's own torso length yields a measure independent of camera distance, since both quantities scale identically with range:

$$e = h_{bb} \,/\, L_t \tag{9}$$

2.5 Joint flexion angles

Flexion at a joint is the angle subtended by the two limb segments meeting at it, obtained from the cosine rule on the aspect-corrected segment vectors:

$$\varphi = \arccos\left[\left(v_1 \cdot v_2 \right) / \left(|v_1|\,|v_2| \right) \right] \tag{10}$$

where the knee angle is formed by the triplet (hip, knee, ankle) and the hip angle by (s_{mid}, hip, knee). A joint is evaluated only when all three constituent keypoints satisfy the confidence gate of Section 1.1.

2.6 Standing-bend guard

A subject bending forward from standing presents a near-horizontal trunk and a compressed bounding box, and is the principal source of false fall alerts. The guard establishes that the legs remain beneath the subject:

$$\left(\text{ankle}_y - \text{hip}_y \right) / L_t \;\geq\; 1.2$$

When no ankle is available at sufficient confidence, a fallback requires both an inverted trunk indicator ≥ 0.30 and a knee-drop ratio ≥ 0.60. A subject satisfying the guard cannot enter the fallen state regardless of trunk angle.

2.7 Spatial context

Furniture regions are defined as polygons in image coordinates. Membership is evaluated by a point-in-polygon test on the hip midpoint, which is the subject's support point:

$$\text{pointPolygonTest}(\, P \,,\, h_{mid} \,) \geq 0 \tag{11}$$

2.8 Fallen-state decision

A frame is classified as fallen if either of two independent paths is satisfied. Two paths are required because a subject falling toward or away from the camera produces little change in trunk angle while collapsing markedly in height, and a subject falling laterally produces the converse.

Path A — horizontal: $\theta > 65° \;\wedge\; \alpha < 0.85 \;\wedge\; \neg$ bend-guard

Path B — collapse: $r_h < 0.50 \;\wedge\; \alpha < 1.20 \;\wedge\; e < 2.5$

2.9 Temporal confirmation and admissibility gates

A fallen classification must hold for 3 consecutive frames before the state is entered (4 frames from a stumble precursor, 2 from a normal walking state). Detections are further excluded when the subject is too small for reliable pose estimation or is truncated by the frame boundary:

reject if $h_{bb} < 0.08 \cdot 360 = 28.8$ px or box within 4 px of the frame border

3. Fall — the angular-velocity discriminant

A fall and a controlled lie-down are indistinguishable by posture alone; they differ in the rate at which the posture is reached. The system therefore computes the peak angular velocity of the trunk over the transition. Trunk angle samples are retained in a time-bounded history and the minimum within a 2.5-second window is taken as the transition onset:

$$\omega = \left(\theta_{last} - \theta_{min} \right) / \left(t_{last} - t_{min} \right) \tag{12}$$

where θ_{min} is the smallest trunk angle recorded within the preceding 2.5 s and t_{min} its timestamp. The history is pruned by elapsed time (30 s) rather than by frame count, so the measure is independent of the achieved frame rate.

$\omega \geq 45$ °/s ⇒ FALL $\omega < 45$ °/s ⇒ LYING DOWN

4. Lying down — positive-evidence gate

Entry into the fallen state by the geometry of Section 2 is necessary but not sufficient to raise a lying-down alert. Positive evidence that the subject is horizontal on the floor must additionally be accumulated. The state is first held for 3.0 s, then evaluated against two mandatory conditions and one of two alternatives.

4.1 Mandatory condition 1 — lower body visible

Both hips and at least one knee must be detected at c ≥ 0.45. A subject whose lower body is not observable cannot be assessed for floor contact, and no alert is raised on inference from the upper body alone.

4.2 Mandatory condition 2 — vertical stack ratio

The vertical distance from the shoulders to the lowest visible leg joint, normalised by torso length, distinguishes a body extended on the floor from one whose mass remains vertically stacked:

$$S = (\max(\text{ankle}_y , \text{knee}_y) - s_{mid,y}) / L_t \tag{13}$$

Observed ranges by posture, from which the admission threshold is set:

Posture	S	Posture	S
Standing	2.5 – 3.0	Seated	1.1 – 1.8
Bending forward	1.4 – 2.0	Lying	0 – 1.0

admission requires $S \le 1.20$

4.3 Silhouette constraint and the two alternatives

box not tall: $\alpha \le 1.60$

and then either:

Alternative A — legs extended: $\max(\varphi_{hip}) \ge 140°$

Alternative B — curled but flat: $\alpha < 0.85 \ \wedge \ e < 2.5 \ \wedge \ \theta > 65°$

Alternative A separates a supine or prone subject, whose hips are near full extension, from a seated subject, whose hips are flexed toward 90°. The larger of the two hip angles is taken, so a subject lying with one leg drawn up is still admitted. Alternative B admits a subject curled on the floor, for whom hip extension is absent but the silhouette is flat and the trunk near-horizontal.

4.4 Evidence persistence

The gate is evaluated per frame and its outcome accumulated, so that a single frame of favourable geometry cannot raise an alert and a single occluded frame cannot suppress one:

alert requires $(\text{passing frames} / \text{evaluated frames}) \ge 0.60 \ \wedge \ \text{passing frames} \ge 3$

4 consecutive failures cancel the accumulation; a rejection is then held for 20 s

5. Prolonged fall — escalation

Time spent on the floor after a fall is the clinically significant quantity. Once a fall is confirmed, elapsed time is measured from the onset of the fall, not from the alert, and escalating notifications are emitted at fixed milestones:

$t \in \{ 30 , 120 , 300 , 900 , 1800 , 3600 \}$ s — maximum 6 notifications per event

6. Inactivity

Inactivity detection is independent of posture classification: it measures whether a tracked subject has ceased to move, whatever position they are in.

6.1 Normalised centroid and smoothing

The bounding-box centroid is normalised to frame dimensions so that the measure is resolution-independent, then smoothed:

$$c_x = x / W_{frame} , \quad c_y = y / H_{frame} \tag{14}$$

$$c_t = 0.30 \cdot c_t^{raw} + 0.70 \cdot c_{t-1} \tag{15}$$

6.2 Stationarity test

Instantaneous speed is computed per frame and the median over a 2-second window is tested. The median is used rather than the mean because a single spurious keypoint displacement would otherwise register as movement and reset the timer:

$$v_i = \sqrt{(\Delta c_x^2 + \Delta c_y^2)} / \Delta t \tag{16}$$

$$\text{median}(v_i)_{2s} < 0.015 \ \Rightarrow \ \text{stationary} \tag{17}$$

6.3 Immobility threshold and escalation

continuous stationarity for 1200 s triggers the first alert

milestones at ×1, ×2, ×4, ×8 of that interval — maximum 4 notifications

6.4 Posture qualifier

Immobility in an abnormal posture is escalated as a distinct alert variant. A subject standing still is exempt, since standing is not a posture from which prolonged immobility carries the same clinical implication:

$\theta > 40° \Rightarrow$ POSTURE variant exempt if $\alpha > 1.5$

7. Agitation — rhythm-or-vigour oscillation test

Agitated behaviour is identified from repetitive hand movement about the head. The discriminating problem is that random tracking noise also produces direction reversals; the test therefore measures not merely that reversals occur but whether their timing is regular.

7.1 Hand-to-head configuration

$\text{wrist}_y < \text{nose}_y \;\wedge\; |\,\text{wrist}_x - \text{nose}_x\,| < 1.0 \cdot w_s$

7.2 Episode continuity

Within a 6.0-second window, only the longest unbroken run of samples is analysed. A gap exceeding 1.5 s breaks the run, and the retained run must span at least 3.0 s. This prevents a rate being computed across disjoint fragments, which would report periodicity that was never observed.

7.3 De-noising

A 3-sample median filter is applied to the wrist trajectory. The unfiltered series is retained in parallel, because the vigour signature of Section 7.6 measures true excursion amplitude, which filtering would attenuate.

7.4 Travel and swing gates

Total excursion is measured as a robust percentile range of the raw vertical wrist position, expressed in shoulder-widths so that the measure is scale-invariant:

$$T = \left(P_{95} - P_{05} \right) \text{ of } \text{wrist}_y^{\text{raw}} \;\geq\; 0.50 \cdot w_s \tag{18}$$

A direction change is counted as a reversal only if the preceding excursion exceeds $0.35 \cdot w_s$; on the unfiltered series the bar is 1.5 × that value.

7.5 Signature 1 — rhythm

Let g_i be the intervals between successive reversals. Regularity is quantified by the coefficient of variation of those intervals:

$$CV = \sigma\left(g_i \right) / \bar{g} \tag{19}$$

A periodic movement yields a low CV, since its intervals cluster about a characteristic period. A memoryless random process yields CV ≈ 1, since its intervals are exponentially distributed with standard deviation equal to the mean. The CV therefore separates genuine repetitive movement from tracking noise that merely produces a comparable count of reversals.

$\text{reversals} \geq 4 \;\wedge\; \text{rate} \geq 0.8\ s^{-1} \;\wedge\; \bar{g} \geq 1.75 \cdot \Delta t$

$CV \leq 0.55$ on the filtered series, or $CV \leq 0.40$ on the raw series

where the constraint $\bar{g} \geq 1.75 \cdot \Delta t$ rejects oscillations whose period approaches the sampling interval, where the reversal count is an artefact of aliasing rather than a measurement.

7.6 Signature 2 — vigour

Movement faster than the frame rate can resolve will fail the rhythm test through aliasing while remaining unmistakably agitated. A second, amplitude-only signature admits such movement:

$T^{\text{raw}} \geq 1.2 \cdot w_s \;\wedge\;$ raw reversals ≥ 4

The oscillation term is satisfied if either signature is satisfied.

7.7 Score composition and sustained alerting

$$A = 0.20 + 0.60 \cdot \Omega \tag{20}$$

where Ω is the oscillation term of Sections 7.5–7.6, equal to 1 when either signature is satisfied and 0 otherwise. The base term of 0.20 applies when both hands are at the head. A single hand contributes 0.15 and elbow elevation 0.25. The base terms alone cannot reach the alert threshold, so a static hand-to-head posture — which is common and not itself agitation — does not alarm.

alert if $A > 0.25$ sustained 15 s, with a 3 s grace period and a 300 s cooldown